Practical No-1

AIM:-Cassandra [datastax Astra].

dashbord

Select -> CQL

1A) Create a TABLE EMPLOYEE and perform operations on it.

```
CREATE TABLE datascience.emp(empID int,deptID int,first_name
varchar,last_name varchar,PRIMARY KEY(empID,deptID));
INSERT INTO datascience.emp(empID,deptID,first_name,last_name)
VALUES(101,11,'Shubham','Mane');
Select * Fromdatascience.emp;
```

Query For DELETING one ROW: -

```
DELETE FROM datascience.emp where empID=105;
Select * Fromdatascience.emp;
```

1B) Create a TABLE STUDENT

Same as 1a

Practical No. 2A

AIM:-Conversion to HORUS format.CSV to HORUS

```
import pandas as pd
from datetime import datetime
df = pd.read_csv("2-.csv", encoding="UTF-8")
print("Original Data:")
print(df)
processdata = df.copy()
processdata.drop("ISO-2-CODE", axis=1, inplace=True)
processdata.drop("ISO-3-Code", axis=1, inplace=True)
print("\nAfter Delet+-ing Columns:")
print(processdata)
new = datetime.now()
dt_string = new.strftime("%d-%m-%y %H:%M:%S")
print("Date and Time:", dt_string)
f = open("CountryCode.txt", "a")
f.write("Deleting column activity recorded: ")
f.write(dt_string)
f.write("\n")
f.close()
processdata.rename(columns={"Country": "CountryName"}, inplace=True)
processdata.rename(columns={"ISO-M49": "Code"}, inplace=True)
processdata.set_index("Code", inplace=True)
processdata.sort_values("CountryName", ascending=False, inplace=True)
print("\nFinal Data:")
print(processdata)
OutputFileName = "Hours_Country_Code.csv"
processdata.to_csv(OutputFileName)
print("\nFile Saved Successfully")
```

make CSV FILE

2.csv using Country,ISO-2-CODE,ISO-3-Code,ISO-M49

Practical No.2B
Aim:CSV to AUDIO
Code

```
from scipy.io import wavfile
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
def show_info(aname,a,r):
print("audio",aname)
print("rate:",r)
print("shape:",a.shape)
print("dtype:",a.dtype)
print("min,max:",a.min(),a.max())
def plot_info(aname,a,r):
stitle='Signal Wave-' + aname + "at" + str(r) +"hz"
plt.title("Signal Wave")
slegend=[]
for i in range(a.shape[1]):
slabel="CH"+str(i+1)
slegend=slegend+[str(i+1)]
plt.plot(a[:,i],label=slabel)
plt.legend(slegend)
plt.show()
wavfile1="sample.wav"
print("Processing:",wavfile1)
inputRate,inputData=wavfile.read(wavfile1)
show_info("2 Channel",inputData,inputRate)
plot_info("2 Channel",inputData,inputRate)
processdata=pd.DataFrame(inputData)
scolumns=["Ch1","Ch2"]
Processing: sample.wav
audio 2 Channel
rate: 44100
shape: (262094, 2)
dtype: int16
min,max: -6038 6056
<ipython-input-6-13c3d552828a>:3: WavFileWarning: Chunk (non-data) not understood, s
inputRate,inputData=wavfile.read(wavfile1)
```

Practical 3
Aim: Remove bad data and wrong formats from the given data

```
import string
import datetime as dt
# 1.Removing leading or trailing spaces from data
print('#1 Removing leading or trailing spaces from a data entry')
baddata = "  Data Science with too many spaces is bad!!!   "
print('>',baddata, '<')
cleandata = baddata.strip()
```

```
print('>',cleandata,'<')
 # 2.Removing non-printable characters from data
print('\n#2 Removing non-printable characters from a data entry')
baddata = "Data\x00Science with\x02 funny characters is \x01bad!!!"
print('Bad Data:',baddata)
printable_chars = set(string.printable)
cleandata = ''.join(filter(lambda x: x in printable_chars, baddata))
print('Clean Data:', cleandata)
 # 3.Reformatting date format
# Convert YYYY-MM-DD to DD Month YYYY
print('\n#3 Reformatting data entry to match specific formatting criteria')
baddate = dt.date(2019, 10, 31)
baddata = baddate.strftime('%Y-%m-%d')
gooddate = dt.datetime.strptime(baddata, '%Y-%m-%d')
gooddata = gooddate.strftime('%d %B %Y')
print('Bad Data:', baddata)
print('Good Data:', gooddata)
```

Practical 4

Aim: Accessing the Data

A) Create a Data Frame using Python

```
import pandas as pd
data = { "name": ["Sanika","Munazza","Pratiksha"]}
df = pd.DataFrame(data)
df.insert(1,"roll no:",[12,13,14])
print(df)
```

B) Insert column, name, age, roll no using Pandas

Same as a just add df.insert for age

c)Remove rows position 3 and after position 5 truncate

```
import pandas as pd
data = { "age": [23,24,25,26,27,28,29]}
df = pd.DataFrame(data)
newdf=df.truncate(before=3,after=5)
print(newdf)
```

d) Extract the name column from the data frame

```
import pandas as pd
data = { "name": ["Sanika","Munazza","Pratiksha"] ,"age":[22,23,24] }
df = pd.DataFrame(data)
print(df.get("name"))
```

Practical 5

AIM:- transforming data car image

Code

```
import matplotlib.pyplot as plt
from PIL import Image
import numpy as np
 img = "cat.jpg"
 imageIn = Image.open(img)
 fig1 = plt.figure(figsize=(10, 5))
```

```
 fig1.suptitle("Ducati", fontsize=25, color="green")
 imgplot = plt.imshow(imageIn)
 plt.axis("off")
 plt.show()
imagewidth, imageheight = imageIn.size
 imageMatrix = np.asarray(imageIn)
 pixelscnt = imagewidth * imageheight
 print("Pixel:", pixelscnt)
print("Size:", imagewidth, "X", imageheight)
print(imageMatrix)
```

Practical 6

Aim:- Perform to deal with Missing Values in Pandas

Code

```
import pandas as pd
 df = pd.read_csv("Stu.csv")
 print(df)
 df.info()
 df.dropna(how="all", axis=1, inplace=True)
print(df)
biomean = df["Bio"].mean()
mathsmean = df["Maths"].mean()
chemmean = df["Chem"].mean()
 df["Bio"] = df["Bio"].fillna(biomean)
df["Maths"] = df["Maths"].fillna(mathsmean)
df["Chem"] = df["Chem"].fillna(chemmean)
print(df)
```